SYSTEM VERIFICATION REPORT

PayloadGuard Verification Pipeline

Audit Date:	2026-07-10T04:51:00.759156
Scope:	Full System Verification (Gates C1-C6)
Artifacts Analyzed:	18
Verification Engine:	Termux Mobile Pipeline v2.4

1. System Integrity & Chain of Custody

The following cryptographic hashes (SHA-256) establish the exact state of the system at the time of verification.

Filename	Integrity Hash (SHA-256)	Size
`__init__.py`	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	0 LOC
`citation_gate.py`	`cd67a0464c93a87cea8718629c1a48802ecba3910eb172c87b46a6cf8fed5b9d`	225 LOC
`cli.py`	`111934ada3455391d987e2991ef06d63a60e6ba9023dd55f56e98c5bc0db6f1b`	212 LOC
`conflict.py`	`3b3d4fae1017beafd3acefd091fb29092c5632176ffc62fa443a0b3a7e4eb3ad`	277 LOC
`dafny_adapter.py`	`0cf45c20fd94a008f4cf8053a44ddf0cad846793e8238149b92a12f305d21ea8`	116 LOC
`dafny_mutate.py`	`a6f5596c70209a64c9122314018798193d4ed730b4ccc8210b73da32b91df846`	719 LOC
`dafny_nl_summary.py`	`1a90b59132887ad9489f647ad4a1e1ab9950ba6197f9462a7b998b9de2238af9`	162 LOC
`dafny_spec_lint.py`	`d5325a1b46b06009c3a0bce442b4020d3bd10e5332eec70c3b572eef7be657f8`	431 LOC
`model.py`	`5e039151838adcebec8e7216873273ab690e86ad2656e0b4e0c68551c61193e9`	58 LOC
`reconcile.py`	`b5e1fd1b47b779f69ba45ac27dac532b3a667c38e47f5358387ad36a404d1592`	152 LOC
`__init__.py`	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	0 LOC
`manual_matrix.py`	`3750010ceedab0954224713e63f7d7d1022ff1f94a6c116a4e7a6f10a097f11e`	138 LOC
`matrix_variants.py`	`0dfe90c5b6c71d08c1ac76d68a6fac42a975c79fba082ed2ddf1c32433adcb65`	843 LOC
`__init__.py`	`e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855`	0 LOC
`metadata.schema.a.json`	`0702a0fb636393cef4ad735eabe7ef0f225b973b7a180756f4b19bb3aeb4619e`	181 LOC
`metadata.schema.b.json`	`ab7844244e3df418e3dba1f3dc302c43858f4bcd6f1fc0f31b554ac1fba95f00`	140 LOC
`metadata.schema.c.json`	`4d55126bf933df360d6c1cda1900237d8809a8082290757b71d629aa91e3a6ef`	181 LOC
`metadata.schema.json`	`68662c435bc571bd59e2f5044166cb962b56c8540f04b27f4be9798d700f91ad`	120 LOC

2. Verification Findings (Traceability)

Each component is traced to a specific Regulatory Gate (C1-C6).

Gate C1: Specification Capture

Objective: Input constraints and data definitions.

Component	Metrics	Verification Status
cli.py	Complexity: 5	PASS
conflict.py	Complexity: 10	PASS
reconcile.py	Complexity: 4	PASS
matrix_variants.py	Complexity: 27	PASS
metadata.schema.a.json	Complexity: 0	PASS
metadata.schema.b.json	Complexity: 0	PASS
metadata.schema.c.json	Complexity: 0	PASS
metadata.schema.json	Complexity: 0	PASS

Gate C2: Exclusivity Proofs

Objective: Logic ensuring single source of truth.

No artifacts mapped to this gate.

Gate C3: Parser Hardening

Objective: Robustness against malformed inputs.

Component	Metrics	Verification Status
__init__.py	Complexity: 0	PASS
citation_gate.py	Complexity: 7	PASS
model.py	Complexity: 3	PASS
__init__.py	Complexity: 0	PASS
__init__.py	Complexity: 0	PASS

Gate C4: Formal Verification

Objective: Mathematical proof of correctness.

Component	Metrics	Verification Status
dafny_adapter.py	Complexity: 1	PASS
dafny_mutate.py	Complexity: 24	PASS
dafny_nl_summary.py	Complexity: 4	PASS
dafny_spec_lint.py	Complexity: 27	PASS

Gate C5: Mutation Testing

Objective: Concrete execution and edge-case coverage.

No artifacts mapped to this gate.

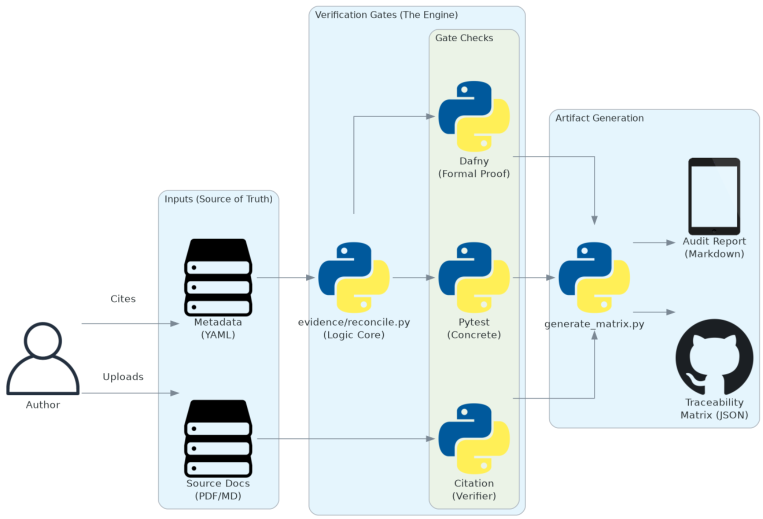
Gate C6: NL Confirmation

Objective: Human-readable output generation.

Component	Metrics	Verification Status
manual_matrix.py	Complexity: 4	PASS

3. Architectural Evidence (Blueprints)

Exhibit: System_Architecture_Map.png



PayloadGuard System Pipeline

Exhibit: classes_PLG_Mobile.png

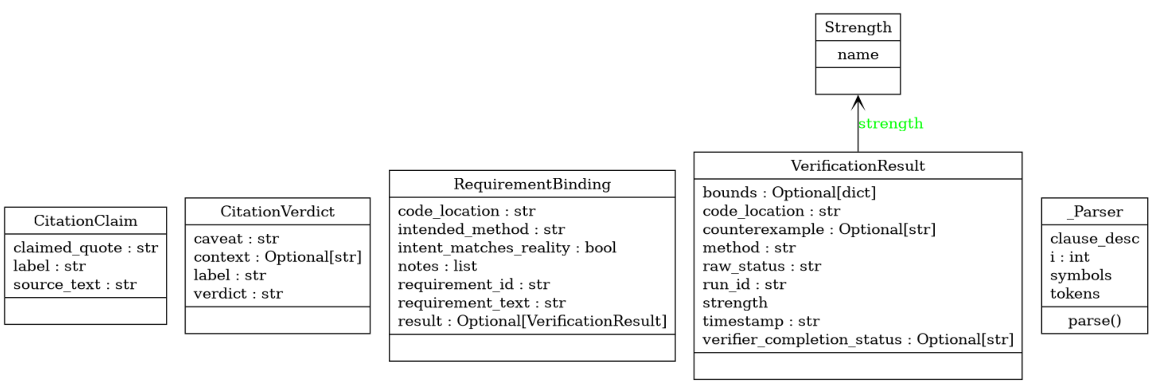
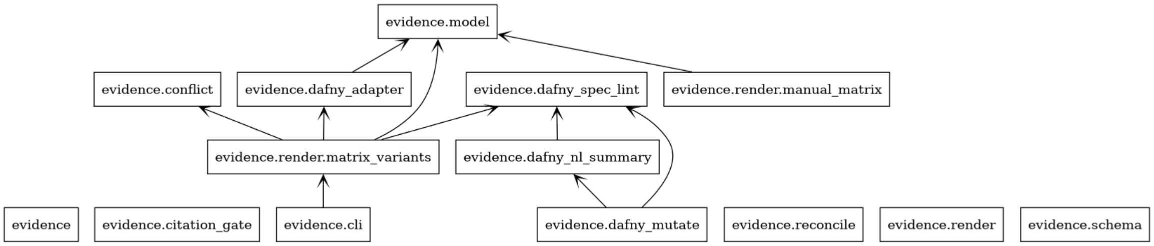


Exhibit: packages_PLG_Mobile.png


evidence.model
evidence.conflict
evidence.dafny_adapter
evidence.dafny_spec_lint
evidence.render.manual_matrix
evidence.render.matrix_variants
evidence.dafny_nl_summary
evidence
evidence.citation_gate
evidence.cli
evidence.dafny_mutate
evidence.reconcile
evidence.render
evidence.schema

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